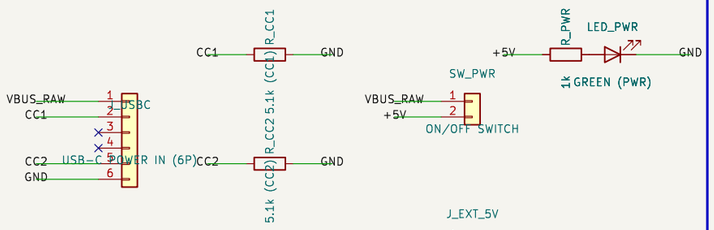
MODULAR CARRIER BOARD v2.0 – SCHEMATIC REVIEW

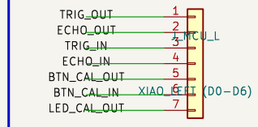
2–S3 Sense | Dual AJ–SR04M Zero–Friction Docks | MAX98357A I2S Audio | Dual Calibration Controls

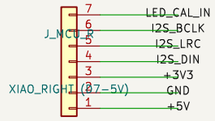
PRIMARY POWER INTAKE & DISTRIBUTION



USB–C with 2x 5.1k CC pull–downs for 5V 3A negotiation.

2. SEEED STUDIO XIAO ESP32–S3 SENSE MCU



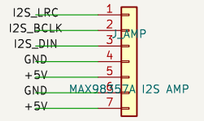


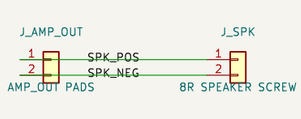
Dual–Core 240MHz ESP32–S3 | 8MB PSRAM

Integrated OV2640 AI Camera Subsystem

Native USB–CDC Telemetry & Flashing

3. I2S CLASS–D AUDIO AMPLIFIER (MAX98357A)



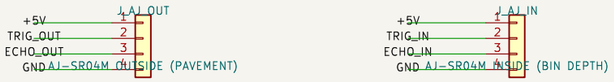


3.2W Class–D Mono DAC/Amp

Direct Digital I2S Audio Bus (Zero Jitter)

2x M2.5 Standoff Mechanical Mounts

–SR04M SENSOR DOCKS (ZERO CABLE FRICTION)

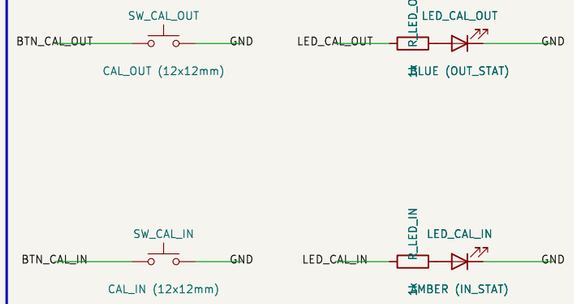


Each dock features 4x M2.5 mechanical standoffs.

Transducer jacks face left PCB edge directly into free air,

guaranteeing ZERO cable bending friction or crowding.

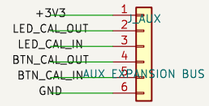
5. SIZABLE 12x12mm CALIBRATION CONTROLS



SW_CAL_OUT: 15–ping pavement reference distance calibration.

SW_CAL_IN: Dynamic container depth calibration (1m to 4m).

6. AUX EXPANSION & SPEC NOTES



BOARD FABRICATION SPECIFICATION:

Dimensions: 135.0 mm x 95.0 mm

Layers: 2–Layer FR4 | 1.6 mm Thickness | 1 oz Cu

Mounting: 4x M3 Corner Standoffs (4.0 mm inset)

Module Docks: 10x M2.5 Standoff Mechanical Holes

Power Input: 5V DC via USB–C or Screw Terminal

Dual AJ–SR04M Docks, Dedicated USB–C Power Entry, 12x12mm Calibration Controls
Master Modular Carrier Board for Seeed Studio XIAO ESP32–S3 Sense

Hochschule Niederrhein (HSNR) – Fachbereich Elektrotechnik & Informatik

Sheet: /
File: Daneben_Prototype_Carrier_Board.kicad_sch

Title: Daneben! Prototype Carrier Board (Modular Showcase Edition v2.0)

Size: A3	Date: 2026–10–26	**Rev: v2.0**
KiCad E.D.A. 9.0.4		Id: 1/1